

INSIGHT

Intelligent In-home Safe Halo Gravity Traction

NC STATE
UNIVERSITY



UNC
CHAPEL HILL

Principal Investigators



Dr. Gregory Buckner
Distinguished Professor in
Mechanical and Aerospace
Engineering at NCSU



Sean Zeller, MBA, MSPO, CPO
Director of Prosthetics and
Orthotics at UNC Hospitals

Lab Manager



Shaphan Jernigan
Lab Manager, NCSU

Undergraduate Researchers



Cole Malinchock



Will Mueller



Pratik Thapa



Edward Katz

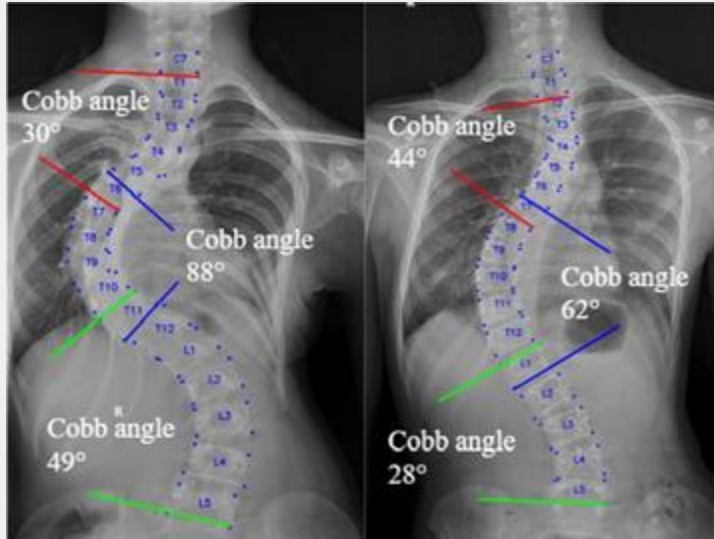


Hailey Nguyen



Emma Brown

Adolescent Idiopathic Scoliosis (AIS)



- **3 million** diagnosis each year
- 10% progress to **severe AIS**
 - **Cobb angle** exceeding 40°
 - Require spinal fusion surgery
- **Halo Gravity Traction (HGT)** is the current pre-surgery treatment

| Existing Pre-Treatment

Halo Gravity Traction (HGT)

- **Non-invasive**, surgically attached halo
- Counterweight traction system
- **3-8 week** treatment, 24 hours a day
- **Improves** surgical outcomes



HGT Setbacks

- Lack of standardization
 - Safety Concerns
 - Accessibility
- Transportability
 - In-patient Cost
 - Patient Compliance





INSIGHT

Intelligent In-home Safe Halo Gravity Traction

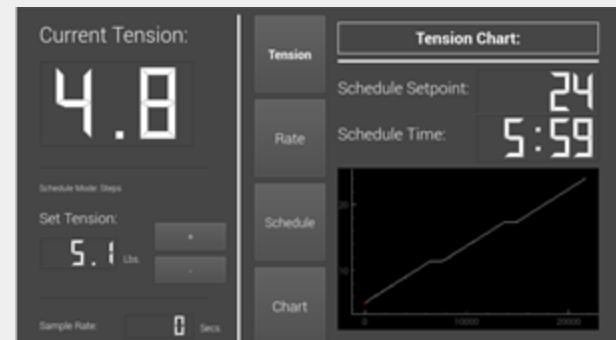
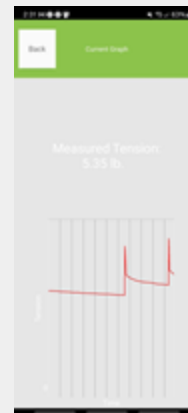
INSIGHT Timeline



June
2023

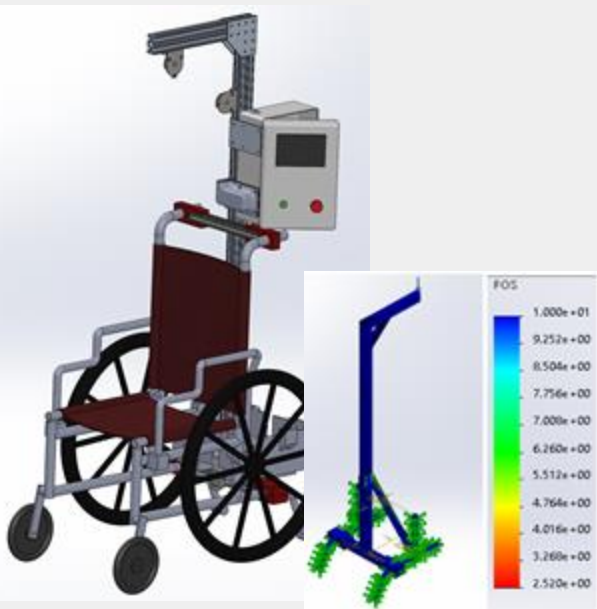


August 2023



December
2023

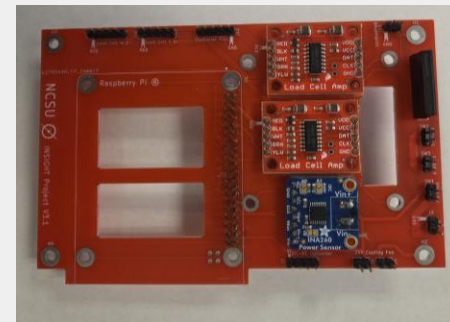
INSIGHT Timeline



January
2024

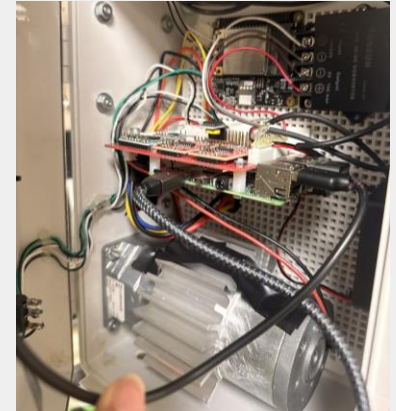
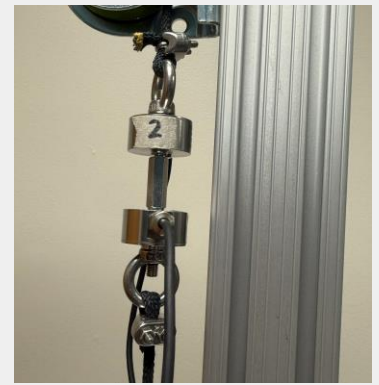
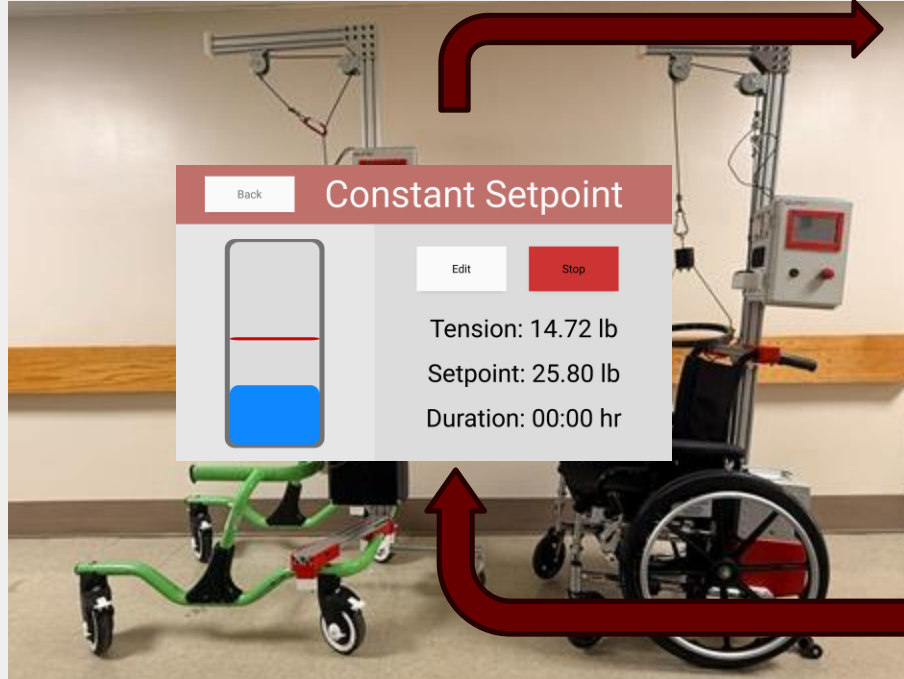


May
2024



August
2024

Current INSIGHT System



INSIGHT Safety Features

E-stops



Sensor Fault-Tolerance



Temperature Sensors



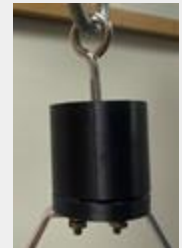
LiFePo Battery



Anti-tipping

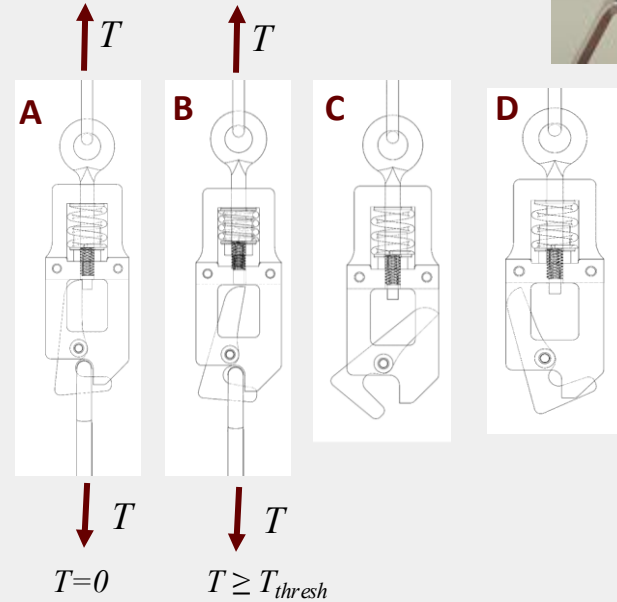


Mechanical Fuse



INSIGHT Safety Features: Mechanic

- 1st Generation: **magnetic**
 - Pros: compact design, repeatable threshold, no mechanical wear
 - Cons: attraction to surrounding objects, non-adjustable threshold
- 2nd generation: novel **mechanical** design
 - Adjustable threshold (20-60 lbf)
 - Resettable with minimal effort
 - Repeatable release tension (± 2 lbf) with static or dynamic loading

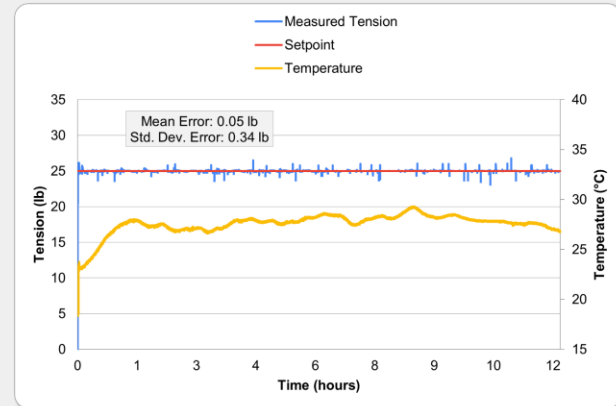
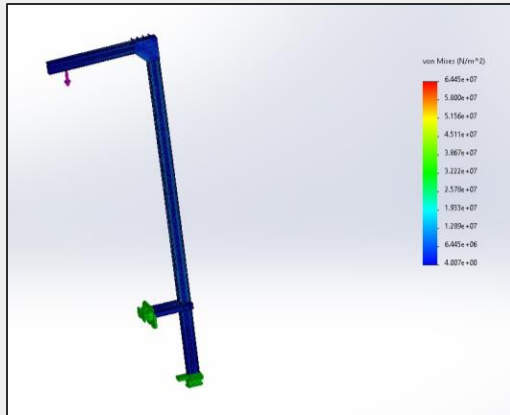


A) zero halo tension, B) tension at threshold, C) halo untethered, D) device reset



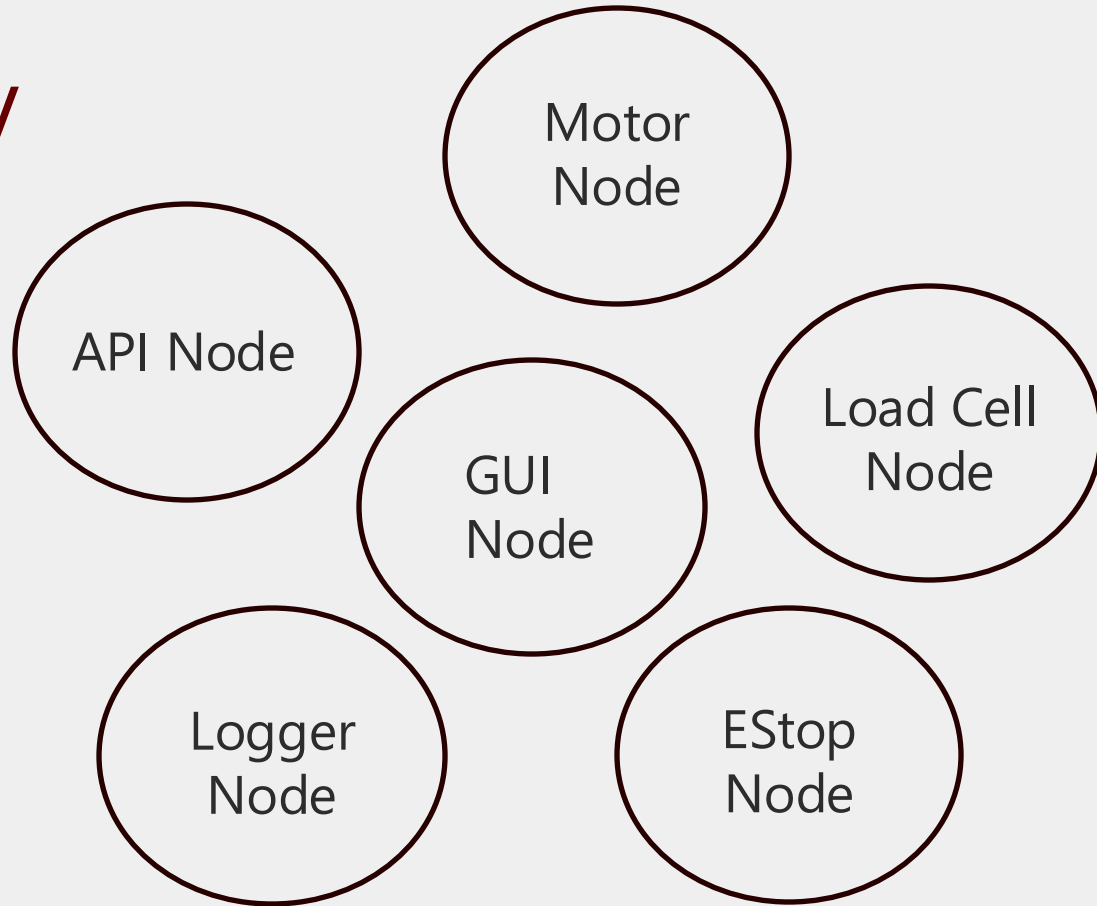
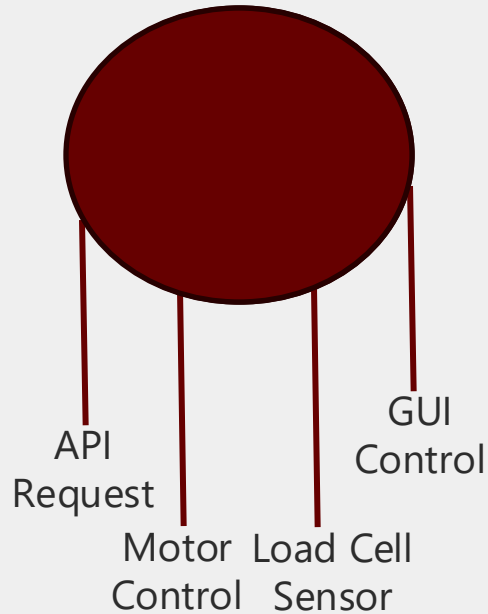
INSIGHT Safety Validation

- Structural Factor of Safety >2.5
 - Utilized SolidWorks Finite Element Analysis
 - IEC 60601-1 Standard
- 12-Hour Endurance Tests
 - Minimal tension error
 - Temperature stayed below 30 °C

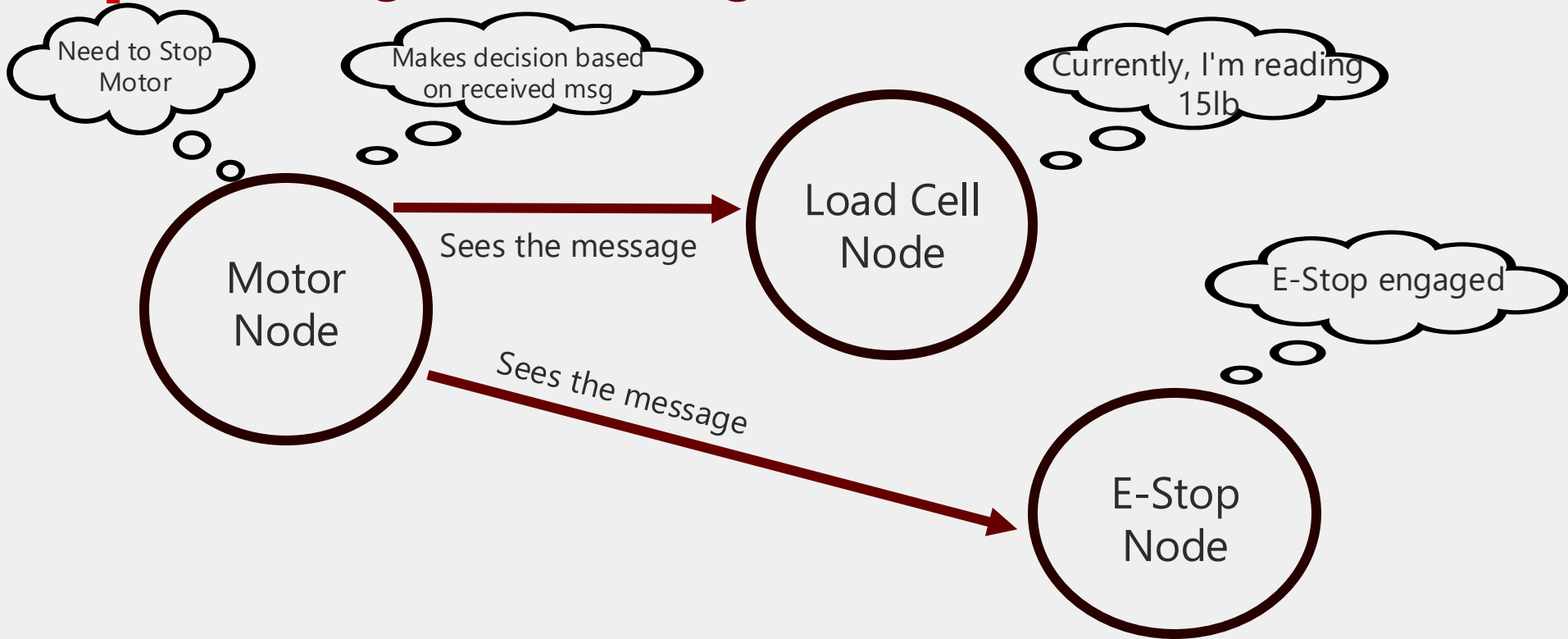


Software Safety

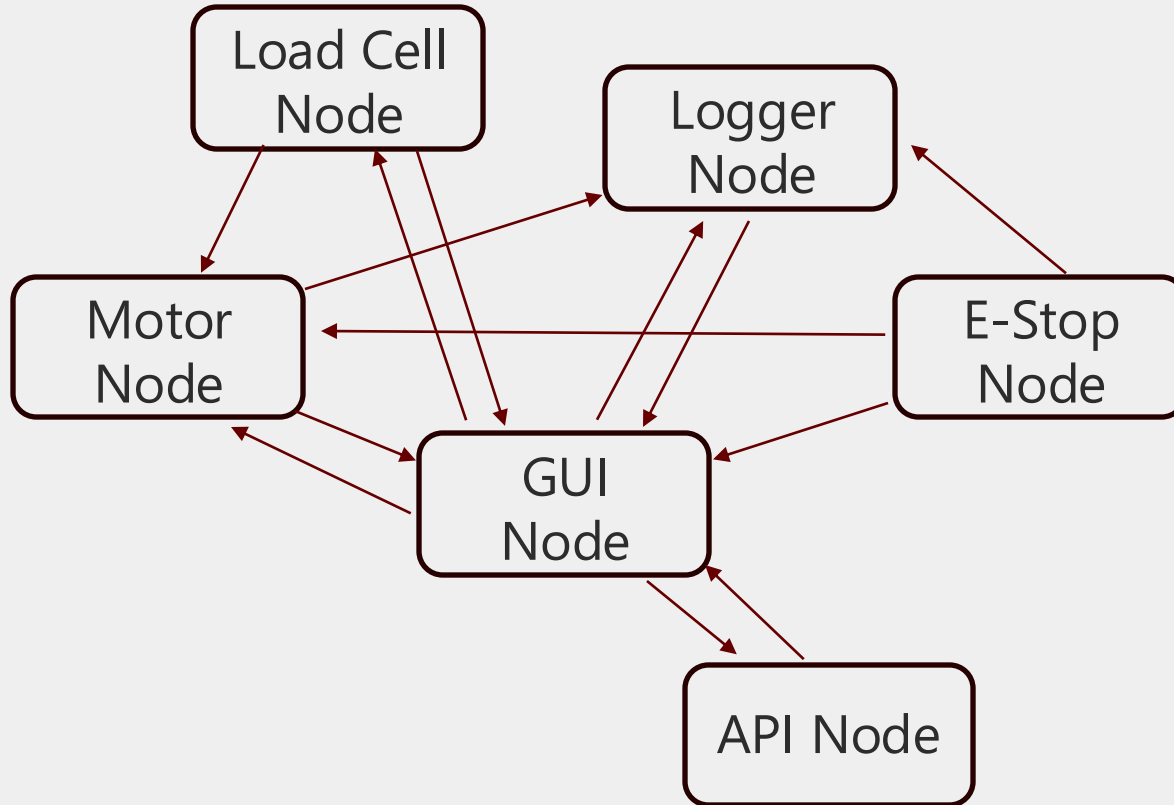
Single Main Process

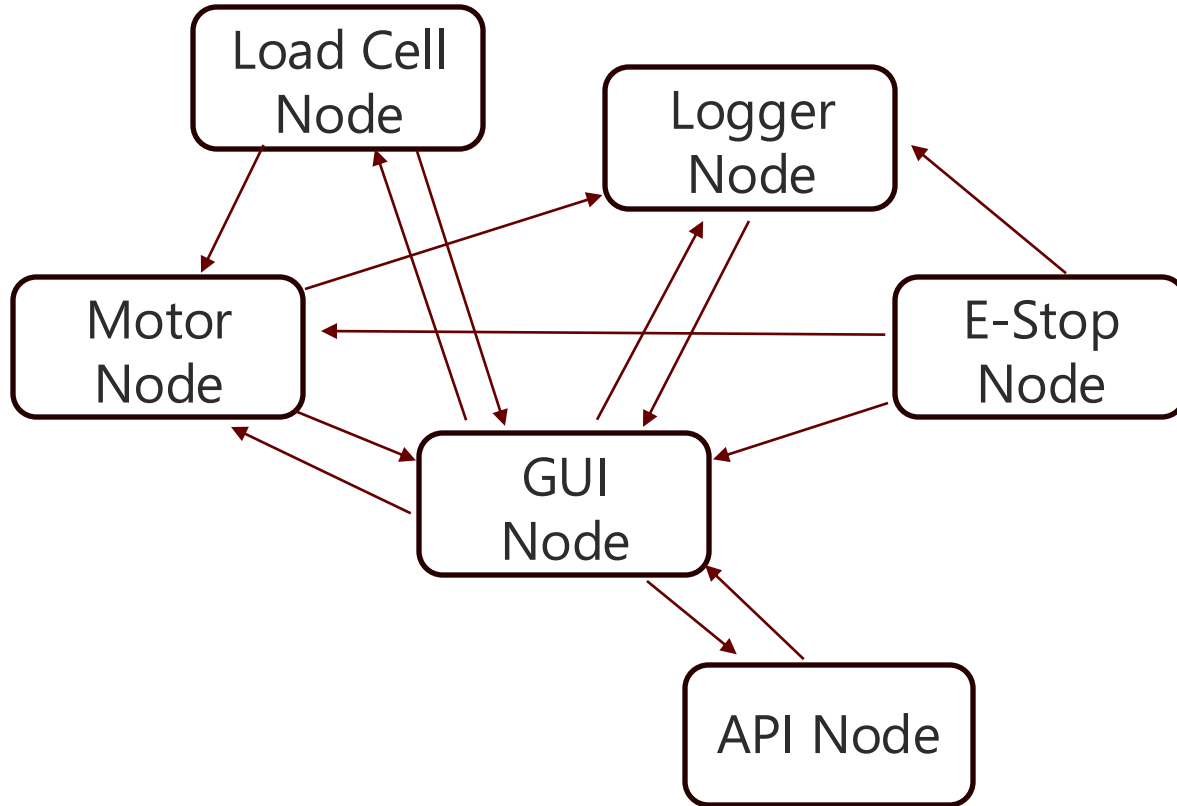


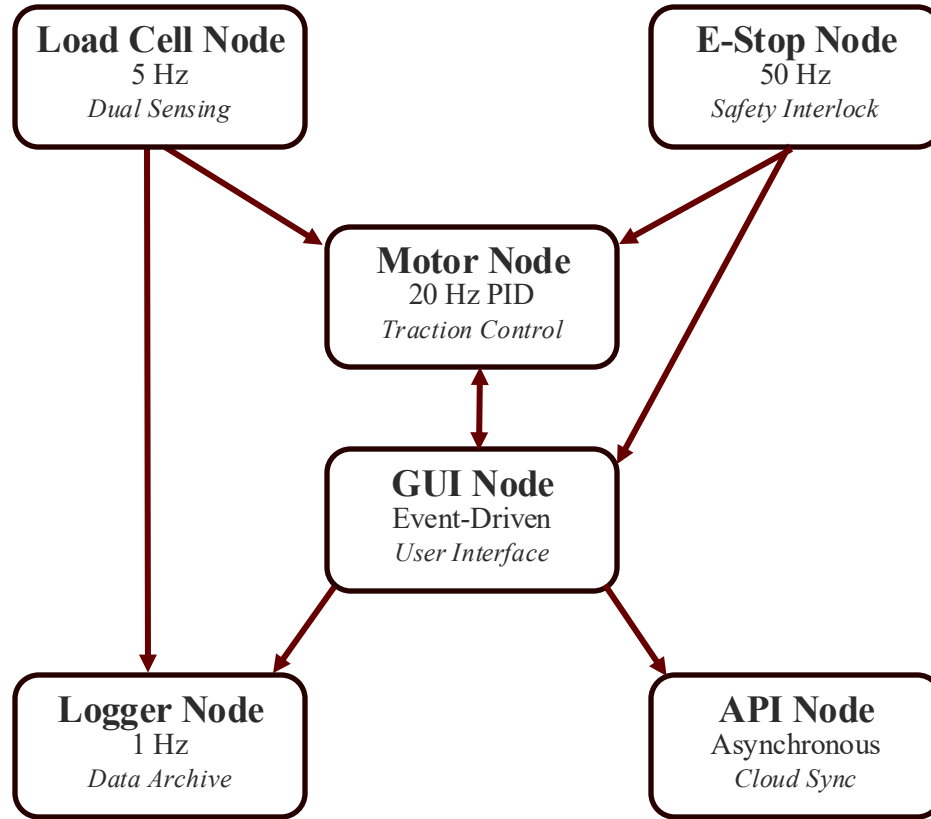
Message Exchange Between Nodes



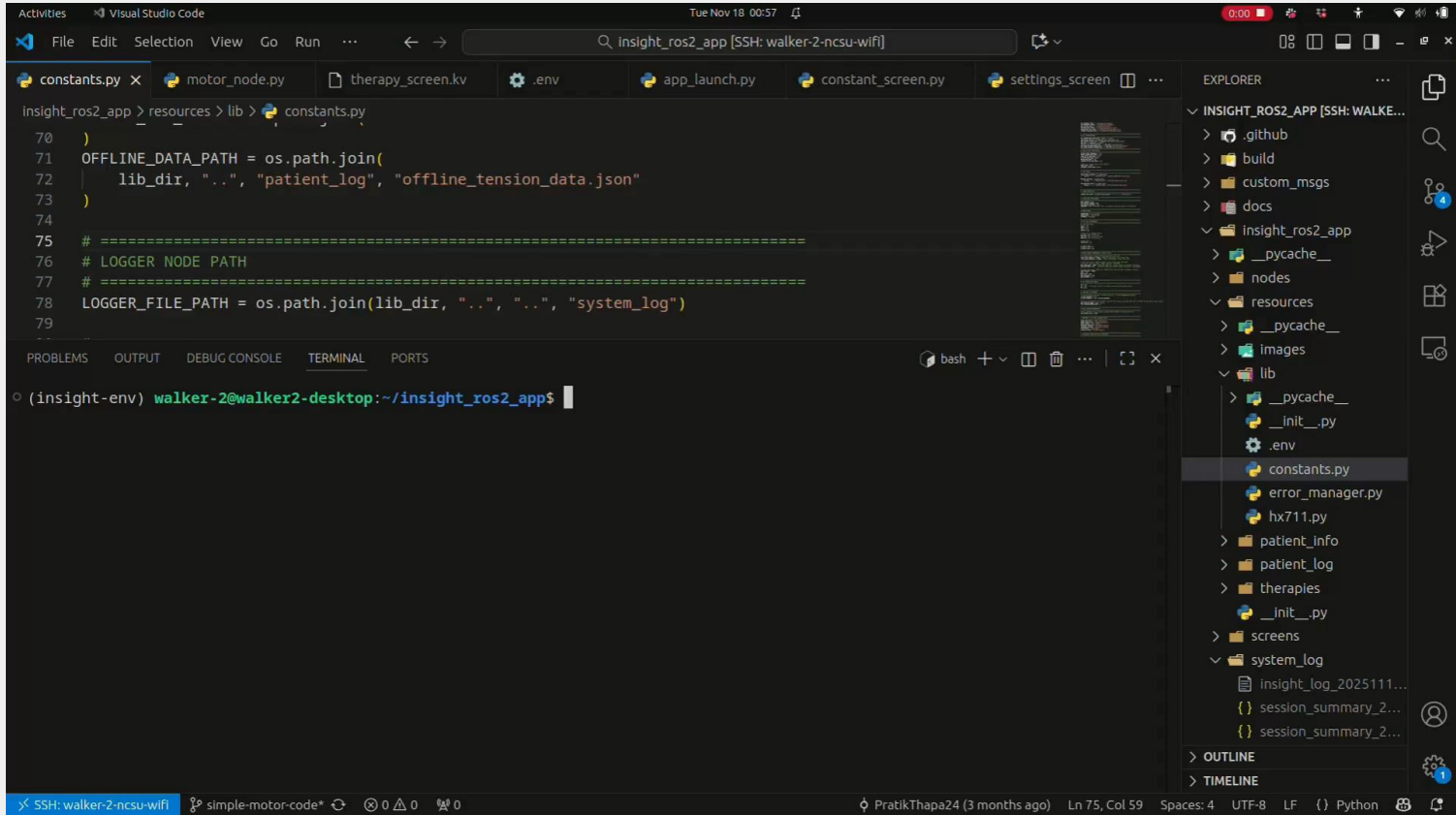
Node Communication







Demo



The screenshot displays the Visual Studio Code interface. The main editor window shows the file `constants.py` with the following content:

```
70 )
71 OFFLINE_DATA_PATH = os.path.join(
72     lib_dir, "..", "patient_log", "offline_tension_data.json"
73 )
74
75 # =====
76 # LOGGER NODE PATH
77 # =====
78 LOGGER_FILE_PATH = os.path.join(lib_dir, "..", "..", "system_log")
79
```

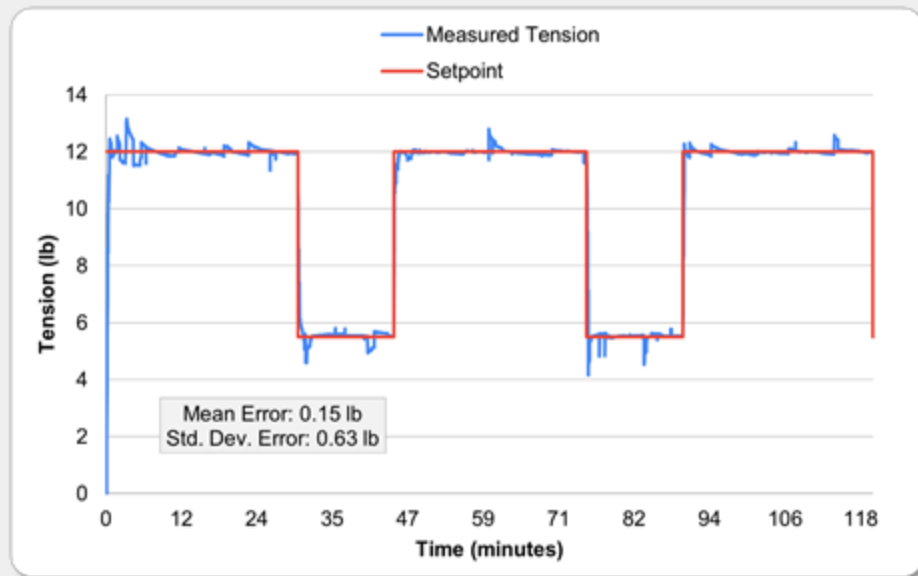
The Explorer panel on the right shows the project structure for `INSIGHT_ROS2_APP [SSH: WALKER-2]`. The `lib` directory is expanded, showing subdirectories like `__pycache__`, `__init__.py`, `.env`, `constants.py` (selected), `error_manager.py`, `hx711.py`, `patient_info`, `patient_log`, `therapies`, `__init__.py`, `screens`, `system_log`, and files like `insight_log_2025111...`, `session_summary_2...`, and `session_summary_2...`.

The Terminal panel at the bottom shows a bash shell prompt: `(insight-env) walker-2@walker2-desktop:~/insight_ros2_app$`.

The status bar at the bottom indicates the file is `simple-motor-code*`, the encoding is `UTF-8`, and the language is `Python`.

INSIGHT Novel Advancements

- **Prescribed therapies** with dynamic traction
 - **High-precision** intervals
 - Preset by physicians
- **Data** storage
 - **Patient compliance**
 - Retrospective studies



INSIGHT Novel Advancements

- **Real-time** remote monitoring
 - Web/Mobile Application
 - Independent cloud communication



INSIGHT Novel Advancements

- **Collapsible** system
 - Improve transportation
 - 4 latches for **both systems**
- Simple, **touch-screen** graphical user interface
 - Input desired tension
 - Displays **real-time** tension



Back

Constant Setpoint

Clear

25.8 lb

1	2	3
4	5	6
7	8	9
0	.	Enter

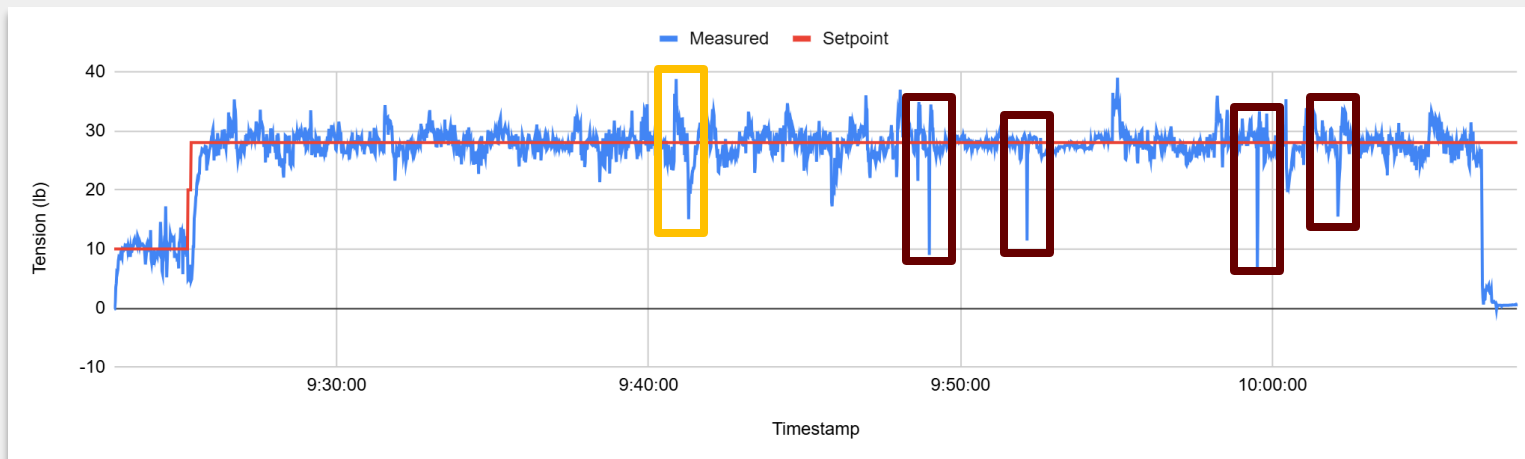
Clinical Trial Progress

3 Patients

- 7-9 years old
- 2/3 completed fusions
- 2/3 also being treated for failure to thrive
- 2-5 months in gravity traction

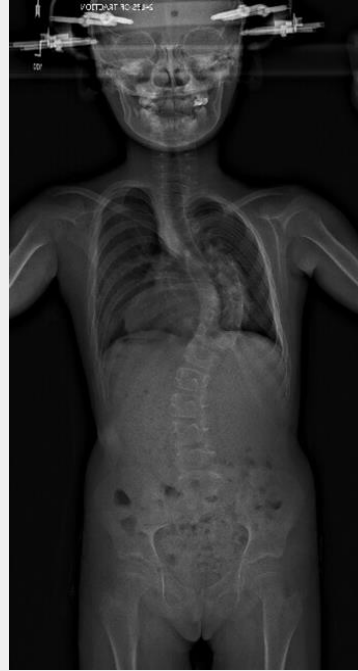


Patient 1

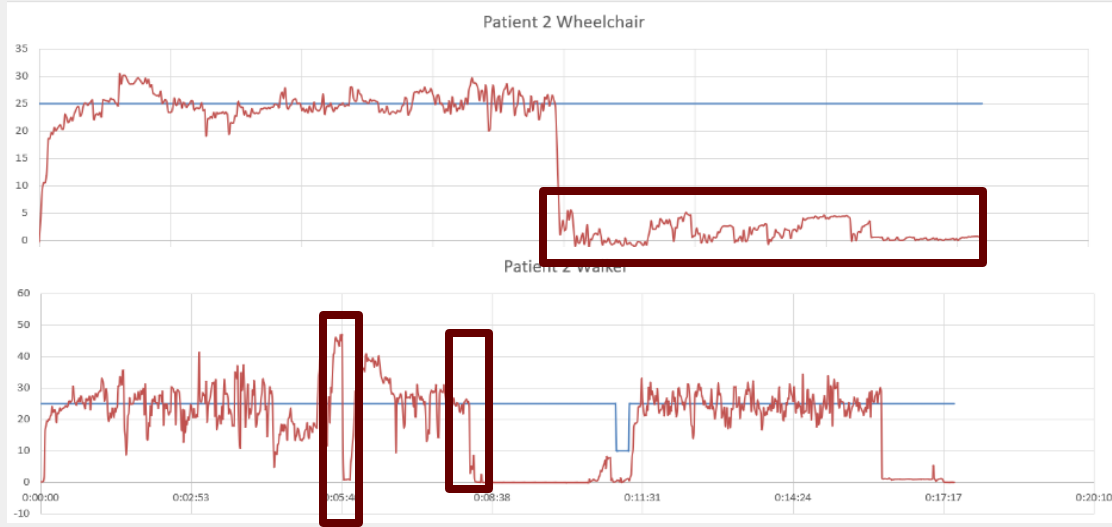


- Accurate traction adjustment to dynamic movement
- Satisfaction with ease-of-use

Patient 1 X-Rays



Patient 2



- Mechanical fuse triggers, accurate real-time tracking
- Limitations of patient motion

| Patient 2 X-Rays



Patient 3



- More limitations of patient motion
- Patient comfort
- How INSIGHT functions with no mechanical fuse



| Economic Plan

Manufacturing Cost:

- Machining (~ \$250)
- Materials (~\$2500)
- General Assembly (~\$800)
- Annual QC & Testing (~\$200)

Total Cost: ~ \$3750

Proposal:

Leasing programs (~\$150/day):

- Generates **3x** the manufacturing cost per year, with **20% profit**
- Covers testing and cleaning between patient use

Current Inpatient Cost: \$4000/day

Outpatient Cost: \$0/day

Comparing Treatments

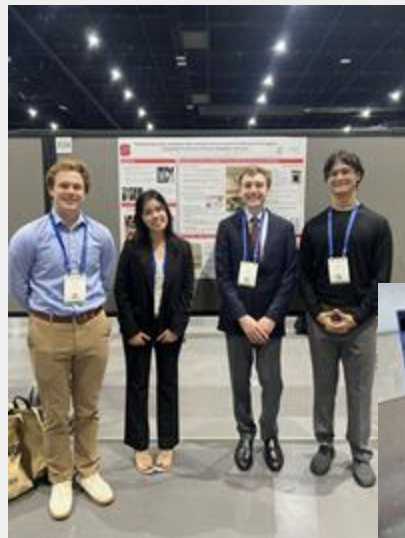
Halo Gravity Traction

- Lack of standardization
- Minimal safety features
- Constant physician monitoring
- Imprecise, static load
- Difficult interface for adjusting

INSIGHT

- Extensive **safety features** and remote monitoring capabilities
- **High precision** and dynamic load features
- Thorough structural testing and improved accessibility
- Transportable and **ease-of-use**
- Effective resource for further research
- **Cost-effective** solution for families

INSIGHT Future



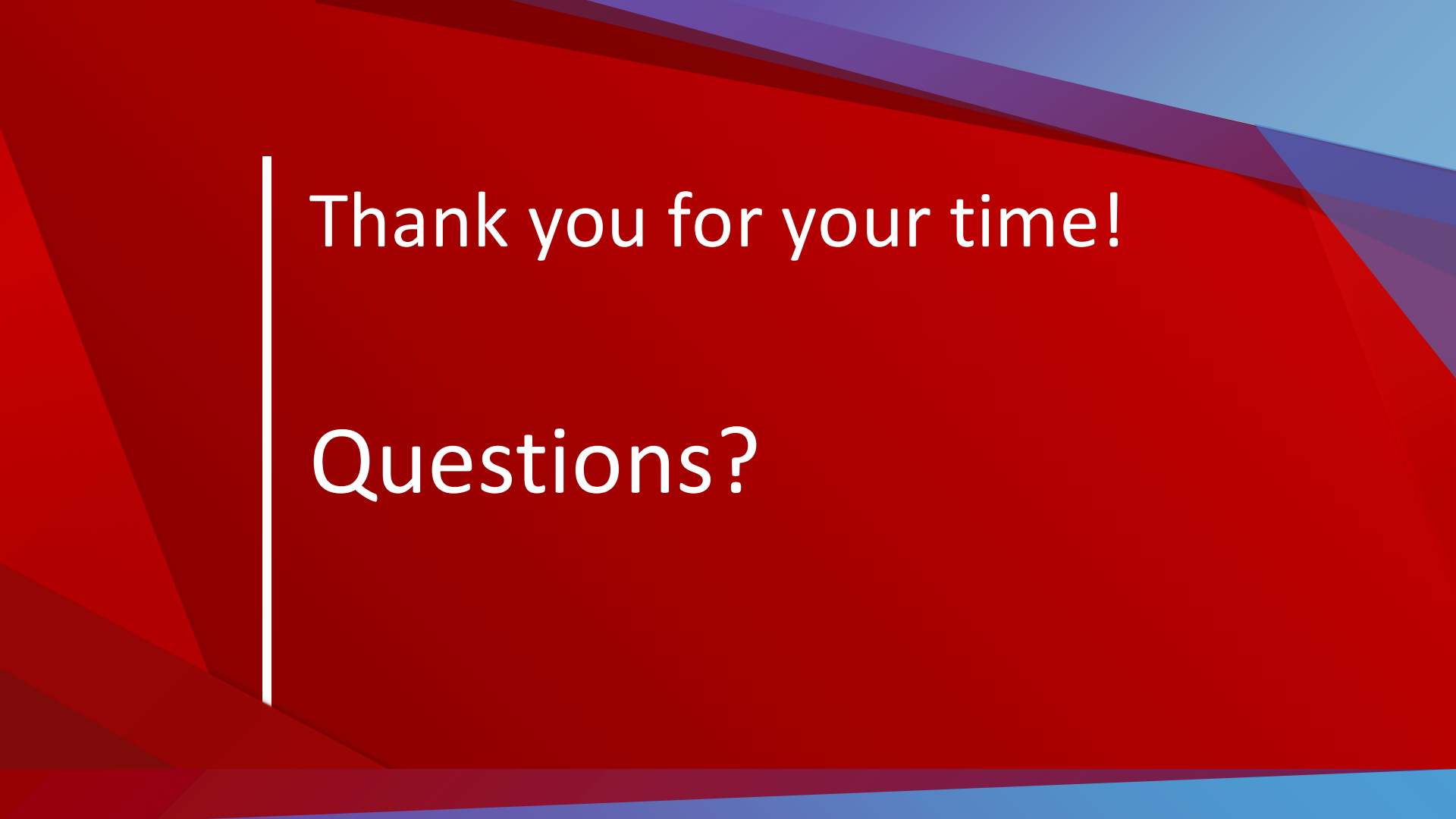
September
2025



December
2025



February
2026



Thank you for your time!

Questions?